



FUTURE INTERNS

SOC&SIEM Log Analysis Using Splunk

CYBER SECURITY

Submitted by:
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Internship**

Repository:
FUTURE_CS_02

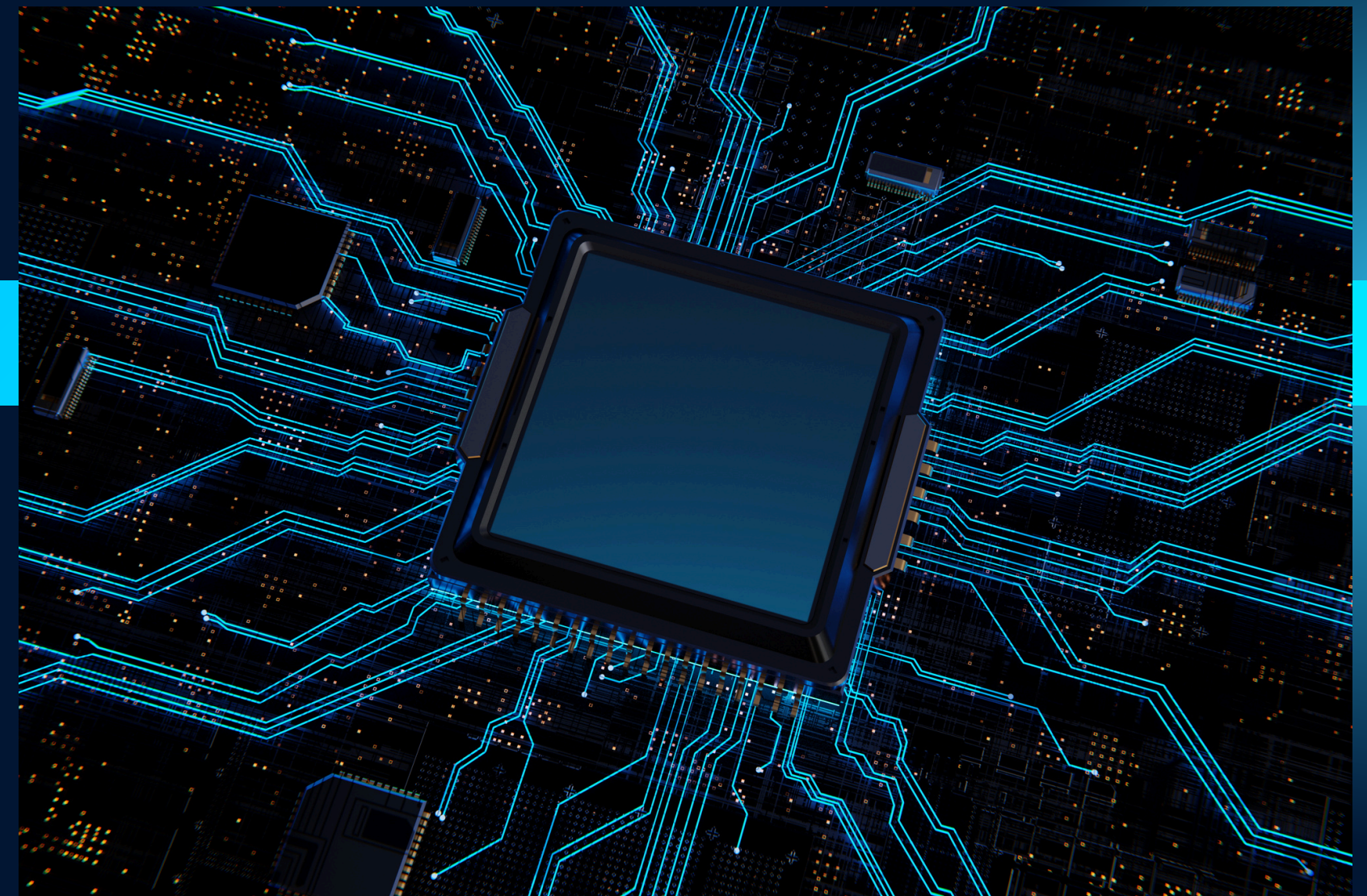
TASK2_REPORT

INTRODUCTION

- This project focuses on analyzing security logs using Splunk SIEM platform.
- The objective is to identify suspicious activities, monitor security alerts, and understand basic incident response procedures.

OBJECTIVE

- Analyze security logs
- Detect suspicious activities
- Identify security alerts
- Perform basic incident response analysis





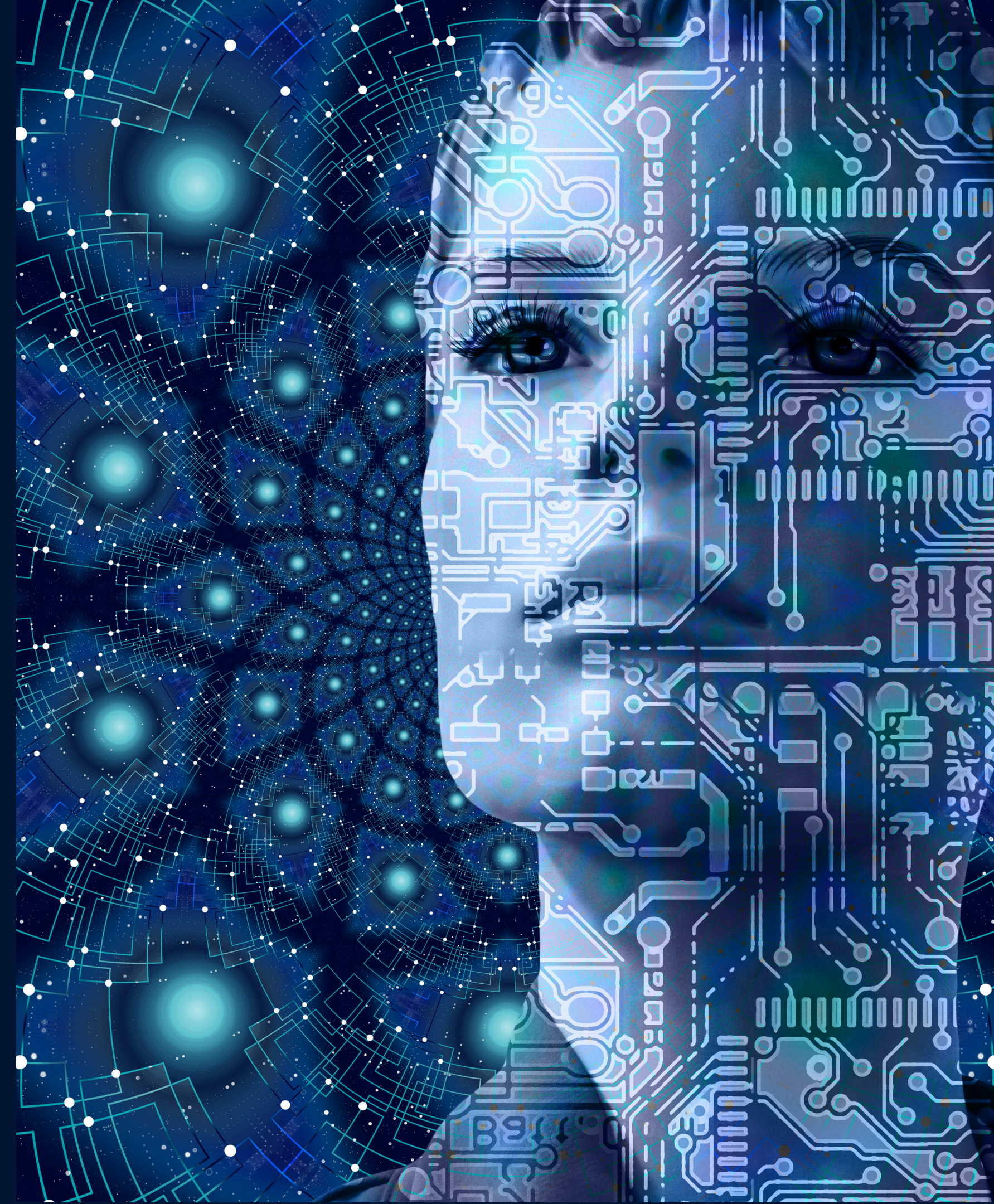
Tools Used

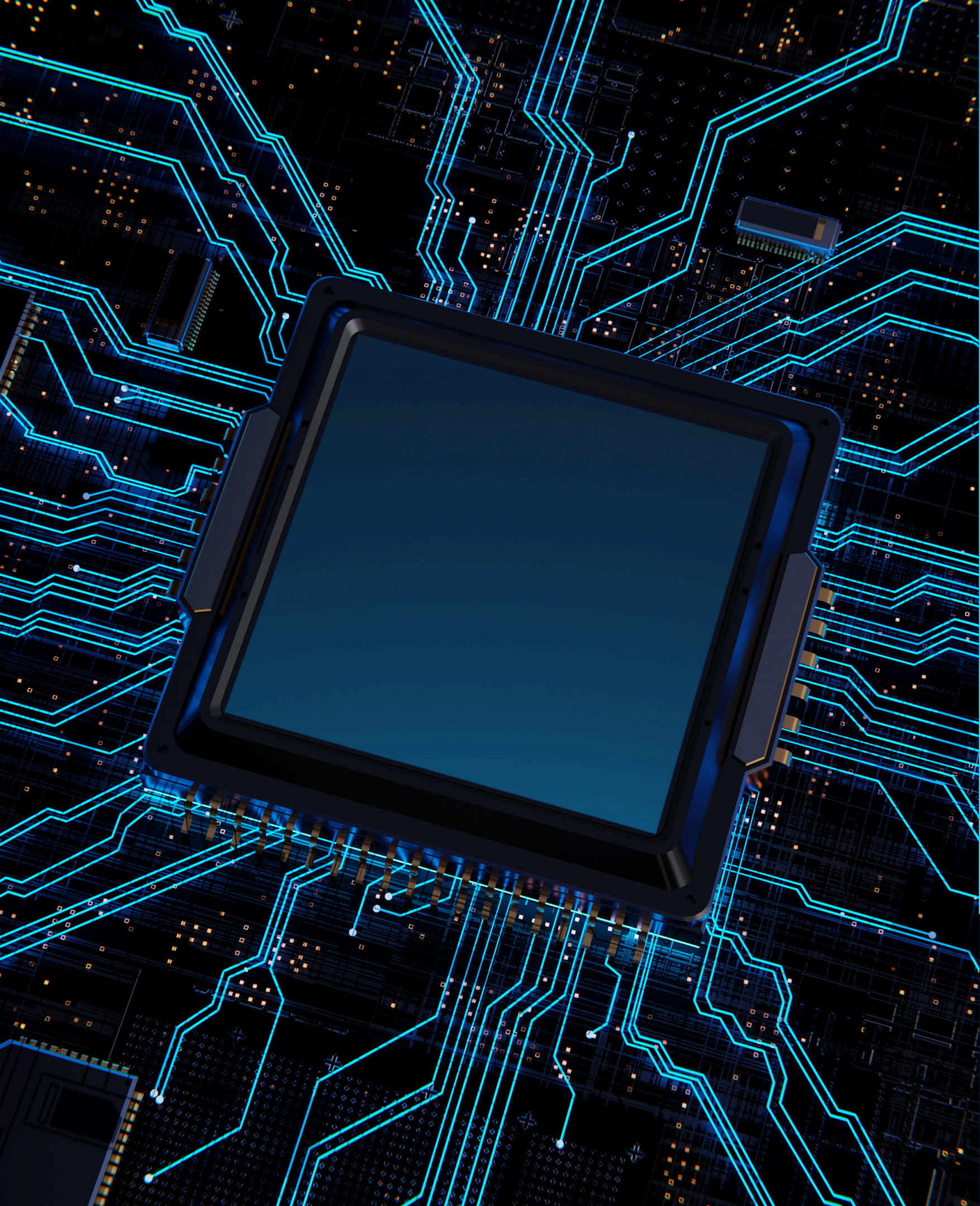
Tool	Purpose
Splunk Enterprise	Log Analysis & SIEM Monitoring
Sample Logs	Security Event Simulation



Log Analysis

The uploaded logs were analyzed using Splunk Enterprise. Multiple suspicious activities such as failed login attempts, malware detection, and unauthorized access attempts were identified during analysis.





DETECTED ALERTS

Alert	Severity	Status
Failed Login Attempts	Medium	Detected
Malware Detection	High	Detected
Unauthorized Access Attempt	High	Detected

whatsapp web - Search

WhatsApp

github - Search

poojithakesanapalli/FU

splunk.com - Search

Thank You for Downloa

Search | Splunk 10.2.3

localhost:8000/en-US/app/search/search?q=search%20source%3D%20sample_logs.txt%20host%3D%20POOJITHA%20sourcetype%3D%20sample_logs&earliest=0&latest=8&sid=1779553199.48&display...

splunk>enterprise

Apps

Administrator

Messages

Settings

Activity

Help

Find

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

Search & Reporting

New Search

Save As

Create Table View

Close

source="sample_logs.txt" host="POOJITHA" sourcetype="sample_logs"

Time range: All time

Q

6 events (before 5/23/26 9:50:00.000 PM)

No Event Sampling

Job

Splunk AI Assistant for SPL is now available.

Events (6)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

+ Zoom to Selection

x Deselect

1 minute per column

Format

Show: 20 Per Page

View: List

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 6

a date_month 1

date_second 6

a date_wday 1

date_year 1

a date_zone 1

a index 1

linecount 1

a punct 5

a splunk_server 1

timeendpos 1

i

Time

Event

>

5/23/26

May 23 10:20:11 Unauthorized admin access attempt

10:20:11.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

>

5/23/26

May 23 10:19:05 Suspicious outbound connection detected

10:19:05.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

>

5/23/26

May 23 10:18:10 Multiple failed login attempts from 45.33.32.156

10:18:10.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

>

5/23/26

May 23 10:17:40 Malware detected on host PC-01

10:17:40.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

>

5/23/26

May 23 10:16:22 Successful login from 192.168.1.15

10:16:22.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

>

5/23/26

May 23 10:15:01 Failed login attempt from 192.168.1.10

10:15:01.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

RESULTS

localhost:8000/en-US/app/search/search?q=search%20failed%20login&earliest=0&latest=8&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=8&sid=1779553337.12

splunk>enterprise

Apps

Administrator

Messages

Settings

Activity

Help

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Datasets

Reports

Alerts

Dashboards

Modules

Search & Reporting

New Search

Save As

Create Table View

Close

failed login

Time range: All time

Q

2 events (before 5/23/26 9:52:18.000 PM)

No Event Sampling

Job

Splunk AI Assistant for SPL is now available.

Events (2)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

+ Zoom to Selection

x Deselect

1 minute per column

Format

Show: 20 Per Page

View: List

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 2

a date_month 1

date_second 2

a date_wday 1

date_year 1

a date_zone 1

a index 1

linecount 1

a punct 2

a splunk_server 1

timeendpos 1

i

Time

Event

>

5/23/26

May 23 10:18:10 Multiple failed login attempts from 45.33.32.156

10:18:10.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

>

5/23/26

May 23 10:15:01 Failed login attempt from 192.168.1.10

10:15:01.000 AM

host = POOJITHA

source = sample_logs.txt

sourcetype = sample_logs

whatsapp web - SearchWhatsAppgithub - Searchpoojithakesanapalli/FU|splunk.com - SearchThank You for DownloSearch | Splunk 10.2.3

localhost:8000/en-US/app/search/search?q=search%20malware&earliest=0&latest=&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=8&sid=1779553465.14

splunk>enterpriseAppsAdministratorMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboardsModulesSearch & Reporting

New Search

malware

Time range: All time

Save AsCreate Table ViewClose

1 event (before 5/23/26 9:54:25.000 PM)No Event SamplingJob

Events (1)PatternsStatisticsVisualization

Timeline formatZoom OutZoom to SelectionDeselect1 millisecond per column

< Hide FieldsAll Fields

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
date_hour 1
date_mday 1
date_minute 1
date_month 1
date_second 1
date_wday 1
date_year 1
date_zone 1
index 1
linecount 1
punct 1
splunk_server 1
timeendpos 1

FormatShow: 20 Per PageView: List

i	Time	Event
>	5/23/26 10:17:40.000 AM	May 23 10:17:40 Malware detected on host PC-01 host = POOJITHA source = sample_logs.txt sourcetype = sample_logs

whatsapp web - SearchWhatsAppgithub - Searchpoojithakesanapalli/FU|splunk.com - SearchThank You for DownloSearch | Splunk 10.2.3

localhost:8000/en-US/app/search/search?q=search%20unauthorized&earliest=0&latest=&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=8&sid=1779554039.17

splunk>enterpriseAppsAdministratorMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboardsModulesSearch & Reporting

New Search

unauthorized

Time range: All time

Save AsCreate Table ViewClose

1 event (before 5/23/26 10:04:00.000 PM)No Event SamplingJob

Events (1)PatternsStatisticsVisualization

Timeline formatZoom OutZoom to SelectionDeselect1 millisecond per column

< Hide FieldsAll Fields

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
date_hour 1
date_mday 1
date_minute 1
date_month 1
date_second 1
date_wday 1
date_year 1
date_zone 1
index 1
linecount 1
punct 1
splunk_server 1
timeendpos 1

FormatShow: 20 Per PageView: List

i	Time	Event
>	5/23/26 10:20:11.000 AM	May 23 10:20:11 Unauthorized admin access attempt host = POOJITHA source = sample_logs.txt sourcetype = sample_logs

whatsapp web - SearchWhatsAppgithub - Searchpoojithakesanapalli/FU|splunk.com - SearchThank You for DownloSearch | Splunk 10.2.3

localhost:8000/en-US/app/search/search?q=search%20source%3D%20sample_logs.txt%20%7C%20stats%20count%20by%20source&earliest=0&latest=&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=8&sid=1779554039.17

splunk>enterpriseAppsAdministratorMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboardsModulesSearch & Reporting

New Search

source="sample_logs.txt" | stats count by source

Time range: All time

Save AsCreate Table ViewClose

6 events (before 5/23/26 10:12:13.000 PM)No Event SamplingJob

EventsPatternsStatisticsVisualization

Chart: Column ChartFormatTrellis

count

sample_logs.txt source

count

6



RECOMMENDATIONS

- Implement strong authentication methods
- Monitor suspicious login attempts
- Restrict unauthorized access
- Perform regular log monitoring
- Maintain updated security policies



CONCLUSION

This project provided practical exposure to SIEM monitoring, log analysis, and incident response using Splunk Enterprise. The analysis helped in understanding basic SOC operations and cybersecurity monitoring techniques.